Soobin Yoon | Curriculum Vitae

University of Minnesota(Twin Cities), Minnesota, USA
Email | Google Scholar | github | LinkedIn

Education

Ph.D. Student in Aerospace Engineering and Mechanics — **Fall 2026–**
University of Minnesota, Twin Cities — *Minneapolis, MN, USA*
Advisor: Prof. Yue Yu

B.S. in Information and Communication Engineering — **Mar. 2021–Feb. 2026**
Dongguk University — *Seoul, Korea*
Degree conferred: Feb. 2026

Research Areas

Optimization and Convex Optimization for Cooperative UAV/UGV Systems; Load Balancing and Path Planning; Vehicle Trajectory Control; and Wireless Communication (IEEE 802.11ax/11be).

Research Experience

Path Optimization for Heterogeneous Autonomous Systems — *Sep. 2026 – Present*
- Conducting research on path optimization and coordination for heterogeneous autonomous systems
- Investigating optimization-based approaches for multi-agent path planning
- Advisor: Prof. Yue Yu, University of Minnesota

UORA Scheduling in IEEE 802.11ax Networks — *Sep. 2025 – Feb. 2026*
- Conducted research on uplink OFDMA random access (UORA) scheduling and developed a MATLAB-based simulator
- First author of a conference paper accepted at the Korean Institute of Communications and Information Sciences (KICS) (poster)
- Advisor: Prof. Eunchan Park, Dongguk Univ.

UAV-UGV Cooperative Load Balancing for Parcel Pickup — *Sep. 2022 – Mar. 2025*
- Designed and analyzed UAV-UGV cooperative scheduling algorithm
- First author of paper published in *IEEE Transactions on Intelligent Transportation Systems (IEEE T-ITS)*
- Advisor: Prof. Kisong Lee, Dongguk Univ.

Publications

Journal Publications

[1]: **S. Yoon**, K. Lee, "Optimal Load Balancing of Cooperative UAV-UGV Parcel Pickup to Minimize Completion Time," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 8, pp. 12712–12718, Aug. 2025. (Impact Factor 2024: 8.4, Q1 journal)

Conference Publications

[2]: **S. Yoon**, E.-C. Park, "Adaptive Access Control for IEEE 802.11ax UORA Based on Per-Station Transmission History," in *Proc. of the Korean Institute of Communications and Information Sciences (KICS)*, 2026. (poster)

Teaching Experience

2026 Fall: TA for Orbital Mechanics (UMN)
2023–2024: TA for Microprocessors, Intro to Programming for ICT, AI Programming, and Info. & Comm. Math. (DGU)

Selected Scholarships & Awards

2026: Excellent Paper Award, KICS Winter General Conference (Undergraduate Paper Competition)
2022–2023: Dongguk Univ. Merit-based Scholarships
2022: Samsung Software Talent Development Scholarship

Technical Skills

Programming: MATLAB, Python, C/C++
Tools: Simulink, CVX, MOSEK
Languages: Korean (Native), English